2

GOVERNANCE FOR WHOM?

Equity, justice, and the politics of sustainable development

In this chapter, we examine issues of equity and justice in the governance of climate change. As we saw in Chapter 1, from concerns about responsibility for causing the problem and the allocation of the burden of responding to it through to questions of climate adaptation, issues of equity and justice run through every aspect of the governance of climate change. Unsurprisingly, therefore, many activists explicitly invoke the language of "climate justice" when campaigning on these issues.[1]

Picking up from Chapter 1, we place the issue of climate change within the broader context of shifting North–South and broader geopolitical relations as well as exploring its links to key development concerns around aid, debt, and trade, and the ways in which efforts to tackle climate change can be made compatible with the alleviation of poverty and the achievement of the UN Sustainable Development Goals (SDGs). This comes amid a growing emphasis on "climate-compatible development": lower carbon models of development that are resilient to the effects of climate change and help lift poorer people out of poverty.[2] As with other chapters in this book, we suggest that because of its deeply interconnected nature, addressing climate change involves a sub-set of wider governance processes which shape and at the same time are affected by climate governance. We look in turn at three key areas where these issues come to the fore: the question of responsibility; the question of who pays for action on mitigation and adaptation; and the question of who bears the costs of actions and inactions on climate change. We conclude by summarizing some of the key governance challenges that confront efforts to address issues of justice and equity in relation to climate change.

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Whose responsibility? Global warming in an unequal world

Historical and contemporary relations of inequity and injustice form a powerful frame by which poorer countries and peoples make sense of climate change as a political issue.[3] Histories of suspicion, distrust, and inequity are played out in the contemporary politics of climate change, accounting for conflicting notions of whom climate governance should serve and how. Whether it is claims about ecological debt or climate justice, there is a powerful sense in which climate change has the potential to further aggravate inequalities within and between societies.[4] This sense of injustice derives from the fact that those who have contributed least to the problem of climate change in the past, including most of the world's poor, are those most susceptible to its worst effects now and in the future. Meanwhile, richer countries are better placed to adapt to the climate impacts that they will suffer. Statements by leaders from developing countries illustrate the strength of feeling on the issue. In 2008, President Museveni of Uganda described climate change as "an act of aggression by the rich against the poor."[5] Meanwhile, at the opening ceremony of the Glasgow COP in 2021, Mia Amor Mottley, Prime Minister of Barbados, asked, "Do some leaders believe they can survive and thrive on their own? Can there be peace and prosperity in one third of the world if two thirds are under siege and facing calamitous threats to their wellbeing?"[6]

Other critics have picked up on the issue of developing countries being reconstituted as a sink for GHG emissions from developed countries through the "flexibility mechanisms," most critically referred to as "carbon colonialism."[7] Even senior government officials within the climate change negotiations such as Raúl Estrada, then chairman of the Ad Hoc Group on the Berlin Mandate (AGBM), noted at the time of the Kyoto agreement:

> My reservation was that the CDM is considered a form of joint implementation, but I don't understand how a commitment can be jointly implemented if only one of the parties involved is committed to limit emissions and the other party is free from a qualitative point of view. Such disparity has been at the root of every colonization since the time of the Greeks.[8]

These concerns have been intensified amid the rush to net zero in the wake of the Paris Agreement with governments and corporations alike seeking to offset their emissions through "nature-based solutions" involving large

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amounts of land and forests in particular environments, often in the global South, leading to concerns about "green grabbing."[9]

The geographies of responsibility have shifted significantly since the early days of the climate regime, as we saw in Chapter 1. While the G77 can still find unity in demanding further action from the developed countries, it contains among its members the world's largest current contributor to climate change (China) and many other countries whose emission trajectories are starting to match those of the United States, the European Union, and Japan (such as India, Brazil, and Mexico). China, the largest contributor, is responsible for 28% of these emissions, followed by the United States (14%), the European Union as a whole (10%), India (7%), Russia (5%), Japan (3%), Korea (2%), Canada (2%), Indonesia (2%), and Iran (2%). All other countries produce about 25% of emissions. Historically, OECD countries emitted the bulk of global GHGs, but the share of BRICS countries contributions to global emissions have been increasing to over 40% since 2010. Emissions of OECD countries peaked in 2007 and have been gradually falling over the past 14 years.[10]

At issue here is the extent to which "differentiated responsibilities" should reflect past, current, or future trends of GHG emissions. This is because much of today's climate change has been caused by emissions generated principally by the development of the global North. In this sense, the North continues to owe a huge "ecological debt" to the South; having consumed a disproportionate share of its global and historical entitlement. A growing number of accounts have shown how the wealth from colonial looting financed growing concentrations of fossil-fueled wealth in colonial powers.[11] Work on carbon debts, meanwhile, demonstrates the extent of the over-use of the commons by richer states and the disproportionate responsibility they bear to now address the impacts of the pollution generated by the wealth they have extracted.[12]

Furthermore, developing countries continue to be home to a significant percentage of the world's poor and their per capita contributions to the problem are dwarfed by those of the United States, for example. The contrasts are stark: the United States, with a population of approximately 323 million, emits 5bn metric tons of carbon dioxide per year compared to a region such as sub-Saharan Africa which emits a combined total of around 823m metric tons of carbon dioxide per year from a population of about a billion people. It becomes clear that the issue of inequity in per capita emissions goes beyond national borders. The Stockholm Environment Institute and Oxfam produced a database that covers 117 countries, around 90% of

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the world's population, and global carbon emissions across the 1990–2015 period which concluded:

> The disproportionate impact of the world's richest people is unmistakable – nearly half of the total growth in absolute emissions was due to the richest 10% (the top two ventiles), with the richest 5% alone contributing over a third (37%). The remaining half was due almost entirely to the contribution of the middle 40% of the global income distribution (the next eight ventiles). The impact of the poorest half (the bottom ten ventiles) of the world's population was practically negligible.[13] (see Figure 2.1)

Taking into account the inequity between and within countries concerning per capita contributions to climate change continues to be a significant challenge for future international agreements and other sites of climate governance.

Beyond the generic framing of common but differentiated responsibility in the UNFCCC, a variety of proposals have been advanced that seek

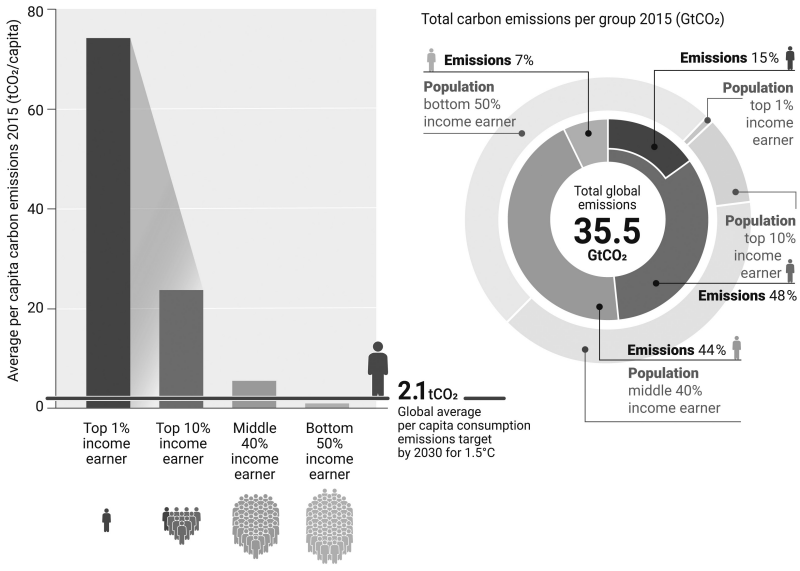


FIGURE 2.1 The polluter elite. Source: United Nations Environment Programme (2020). *Emissions Gap Report 2020*. Nairobi.

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to tackle the thorny issue of per capita emissions and to recognize and reconcile the right to development with the need to dramatically reduce emissions. "Contraction and Convergence" is one such proposal, developed by a small London-based NGO called the Global Commons Institute. The basic idea is that developed countries have to contract their emissions down toward an agreed level which would address the UNFCCC's aim of avoiding dangerous interference in the climate system, while poorer developing countries would be allowed to increase their emissions up to the agreed level. Per capita entitlements would then be bought and sold to harness the market toward this climate stabilization goal. The idea has many opponents, not least the United States and other larger polluters who insist that permits would have to be allocated based on existing need (a grandfathering system) rather than a per capita basis which would favor developing countries. The idea that everyone has equal access to the global commons has powerful political and moral purchase, however.

Another proposal is the Greenhouse Development Rights framework developed by the Stockholm Environment Institute, Christian Aid, and the Heinrich Böll Foundation. It aims to promote the right to development in a carbon-constrained world while holding warming below 2°C, a task which may already seem impossible.[14] Countries' abatement targets would be calculated by the number of people above a development threshold (US$20 a day): the "global consuming class" as they refer to it. Responsibility and capacity indicators (gross domestic product (GDP), population, cumulative emissions), including income available after meeting basic needs, are used in the calculations. Such an approach helps to go beyond aggregated national figures of wealth and development, challenging divisions between "developed" and "developing" countries, while also providing a quantitative handle on how to assess obligations to tackle climate change. Implementation of such an approach could be through taxation into a global fund for redistribution for mitigation and adaptation, or via national mitigation targets. Questions remain about whether this proposal is politically realistic, and as such it does not provide a way of bypassing conflicts over how money would be redistributed or by whom, or how capacity and responsibility indicators would be agreed. Nonetheless, such proposals show that it is possible to conceive of responsibility for climate change in different ways, with significant implications for the development of climate policy and its ability to address issues of equity.

In addition to the challenges of historical and future emissions, nation-based or per capita emissions allocations, a further concern is where along global supply networks responsibility for emissions should be

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allocated—whether the point of production or consumption is the appropriate focus.[15] Many rapidly industrializing countries such as China claim their emissions profile is a by-product of demand for low-cost manufactured goods in developed countries. In other words, they should not be penalized for the fact that many countries are able to outsource the most carbon- and energy-intensive stages of the production process to parts of the world where it is cheapest to do so. Accounting for emissions is complicated by how to capture both the carbon embedded in products through their lifecycle and "virtual carbon" from goods and services that are traded around the world. For example, while the UK's own carbon emissions fell by more than 40% between 1990 and 2016, the emissions associated with imports to the UK, including international travel, rose from about 316m tons of carbon in 1990 to 360m tons in 2016.[16]

If climate politics could ever have been characterized as at core a conflict between North and South over responsibility, the rise of these issues on the agenda means that the picture is certainly more complex now. The issues around which developing countries once mobilized have changed, as the political and economic might of those who once relied upon the solidarity of other states as part of the G77 coalition has increased. BRICS (Brazil, Russia, India, China, and South Africa) countries have increasingly accepted commitments to control their emissions of GHGs as part of their Nationally Determined Contributions under the Paris Agreement. Likewise, growing reliance upon flexible mechanisms and market-based instruments has served to fragment the opposition of developing countries to involving themselves in emission reduction projects. Given this diversity, beyond generic calls for OECD countries to show greater leadership, it is unsurprising that speaking as a united bloc makes increasingly less sense for developing countries in climate negotiations. Some of the more powerful governments have been drawn into alternative governance forums for reaching agreements on climate change. Over time, these have included the Asia-Pacific Partnership on Clean Development and Climate, the Renewable Energy and Energy Efficiency Partnership (REEEP), the Australia–China Partnership, the EU–China Partnership on Climate Change, the US–India bilateral agreement on nuclear energy, and agreements on EU–India energy cooperation. We have also seen, however, the proliferation of "climate clubs"[17] such as the Beyond Oil and Gas Alliance (BOGA) led by Costa Rica and Denmark or the Powering Past Coal Alliance which brings together over 48 countries from all regions of the world to support transitions away from coal.[18] At the Glasgow COP, a series of "breakthroughs" were also announced: actions across five key

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economic sectors that together represent more than 50% of global emissions. These set goals to dramatically accelerate the innovation and deployment of clean technologies in five key sectors of the economy by 2030 including power, road transport, steel, hydrogen, and agriculture.[19] These developments have been conceptualized as a form of "minilateralism,"[20] which, while helping to plug certain gaps in the climate governance landscape, also reinforces its fragmentation and polycentric nature. The new geographies of responsibility for climate change are, therefore, not only shaping the trajectory of current and future international climate agreements, but can be seen as one of the factors shaping the emergence of a whole range of new governance arenas for addressing climate change, a topic to which we return in Chapter 3.

Who pays?

A second key issue that has shaped the North–South politics of climate change is the concern about who should meet the costs of addressing the issue. Agreeing on a fair and equitable allocation of responsibility for acting on climate change requires subsequent agreement about who pays for such actions and how. In governance terms, some of the key conflicts are about which institutions should oversee such flows of funds from North to South, and on what terms. As we saw in Chapter 1, the early battle over whether the GEF should be the body to oversee aid and technology transfer reflected concerns about the role of the World Bank with whom many developing countries have had, and continue to have, difficult relations. We also noted how this contest was replayed again over the role of the World Bank in managing the Green Climate Fund (GCF). A further concern, illustrating how broader North–South politics have shaped climate governance, is where the finance for climate change will come from. Reflecting a deep-seated fear that existing aid budgets might be diverted to fund climate change measures, the obligations for Annex 1 parties explicitly call on them to provide "new and additional" financial resources, though the extent to which this has been the case is hotly contested amid controversy about which types of funding count and questionable claims about contributions made.[21]

What is clear is that richer countries are still falling short on their promise to provide US$100 billion per year through the GCF. Data from the Organisation for Economic Co-operation and Development (OECD) found that climate finance reached only US$80 billion in 2019.[22] Of further concern is the fact that the majority of pledged finance is being given

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as loans, with only 20% awarded as grants, adding to the debt burden of poor nations. Poorer countries are also aggrieved about the imbalance in GCF funding toward mitigation and not adaptation to climate change which is a more pressing concern for those countries on the frontline of climate impacts, despite the institutions' aim to split funding 50:50 between mitigation and adaptation. This is significant because UNEP estimates that annual adaptation costs in developing countries are expected to reach US$140–300 billion in 2030 and US$280–500 billion in 2050.[23] In addition, there is now the pressing question of how to raise money for the loss and damage fund agreed at COP27.

Turning first to questions of institutional oversight, it is clear that multilateral and regional development banks have sought to define for themselves a key role in financing mitigation and adaptation based on a recognition that climate change can reverse development gains in so many other areas where they work. A key actor here is the World Bank. In addition to being an implementing agency of the GEF, the financing agency for the climate agreements, the World Bank has been involved in a range of climate change initiatives. Though there are now more than 20 different bilateral and multilateral funds for climate change, the Climate Investment Funds (CIFs) of the World Bank represent more public finance than has ever before been dedicated to climate change. A key role the World Bank seeks to perform is the provision of incentives for the private sector to carry some of the burdens of funding climate protection. Piloting and demonstrating sustainable energy projects help to minimize some of the risks that deter private actors from investing in the poorest areas of the world. For example, the Scaling-Up Renewable Energy Program was launched at the Copenhagen climate summit in 2009 with the aim of catalyzing new economic opportunities for private sector investment in renewable energy production in low-income countries, while the World Bank's Clean Technology Fund identifies its niche in terms of providing finance to developing countries at more concessional rates than the standard multilateral development banks' terms, and at the scale necessary to help provide them with incentives to integrate low-carbon strategies into their development plans and investment decisions.[24]

However, while some parts of the World Bank are promoting action on climate change, many other policies pursued by the same institution do not take their impact on climate change into account. For example, the World Bank still provided over US$9 billion in public finance for fossil fuel projects in 2016, with the vast majority of transactions approved after the Paris Agreement was reached. Indeed, the total MDB finance for oil and gas

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exploration more than doubled from 2015 to 2016, from US$1.05 billion to US$2.15 billion. The World Bank Group, the European Investment Bank, and the Asian Development Bank were the largest financiers of fossil fuels in 2016. At the same time, renewable energy still made up less than a third of MDB energy finance.[25] On this basis, it is clear that climate change has yet to be mainstreamed throughout the activities of the World Bank.

Whatever their limitations may be, development banks do have a key role to play in supporting clean energy transitions in lower-income countries. Energy is clearly pivotal to development and meeting the development needs of the majority of the world's people in a carbon-constrained world presents a global challenge of staggering proportions. Today, 1.6 billion people are without electricity (13% of the world's population), 3 billion people (40% of the world's population) remain without access to clean cooking, and per capita electricity consumption varies more than 100-fold across the world. According to the IEA, the majority of the increase in electricity demand is expected to come from the Asia-Pacific region, primarily China and India in the coming years.[26] Fatih Birol, IEA's Executive Director, noted in 2021

> Emissions from electricity need to decline by 55% by 2030 to meet our Net Zero Emissions by 2050 Scenario, but in the absence of major policy action from governments, those emissions are set to remain around the same level for the next three years.[27]

Without a significant change, of course, most of this will be fossil fuel-based electricity production that will exacerbate climate change. Effective governance of public and private investment in the energy sector will clearly be key to the success of any efforts at climate change governance.

In this context, the second issue concerning who pays is how to generate new sources of funding given the scale of new finance required. The IPCC estimates that 48.75 trillion over the next 15 years will be necessary to limit warming to 1.5°C.[28] There is increasing emphasis upon the need to expand the amount of private investment in addressing climate change in particular, including via carbon markets. Indeed, the Adaptation Fund is in large part financed by sales of certified emission reductions under the UN's Clean Development Mechanism. The share of proceeds amounts to 2% of the value of certified emission reductions issued each year for CDM projects. Since 2010, the Adaptation Fund has committed more than US$ 850 million for climate change adaptation and resilience projects and

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programs, including more than 123 projects which it claims have benefited 28 million people.[29]

In approaches to financing climate change mitigation, growing attention is being paid to the need to capture co-benefits in developing countries: projects that deliver climate and development dividends simultaneously. In line with this argument, the CDM is often highlighted as a great development opportunity for developing countries to use climate-related funds to invest in projects and technologies that benefit the poor. The projects it approves are meant to capture social and environmental benefits as well as deliver emission reductions additional to those that would have been funded anyway. However, in reality, investment has tended to flow to larger developing countries such as China, India, and Brazil (China alone is home to 57.6% of all registered CDM projects)[30] and not, for example, to Africa, where just over 2% of projects are registered. There has also been neglect of broader sustainable development benefits flowing from the projects even if they are successful at reducing the growth of GHG emissions, partly because these benefits are not rewarded in the same way emission reductions are monetarized through the allocation of Certified Emissions Reductions.[31] In governance terms, there are also issues of limited capacity to manage project demand in some countries, as well as to oversee and implement the projects in a way that protects not just the interests of the investor but also the communities which host these projects.[32] We discuss these issues in further depth in Chapters 5 and 6. For the moment, it is sufficient to note that the debate concerning where additional funds for addressing climate change should come from is far from being resolved in terms of either policy or practice, with significant concern that addressing climate change may serve to divert much-needed resources away from other environment and development agendas, will be tokenistic at best, and may serve to promote particular solutions to the climate change problem favored by the developed world rather than those developing countries in receipt of climate finance.

There is also the danger of trying to create islands of formal climate governance in a sea of unregulated, ungoverned flows of trade and finance unguided by the imperative of addressing climate change. For example, despite their global spread, Emissions Trading Schemes (ETSs) still cover just 8% of global GHG emissions.[33] Moreover, only a small percentage of trade, aid, production, and finance is governed by public bodies charged with tackling climate change.[34] Official Development Assistance (ODA) funds are currently just over 1% of foreign direct investment globally.[35] What this highlights then is the pressing need to identify policies,

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strategies, and interventions which are able to steer financial flows, public and private, to where they are most needed but in ways that are consistent with the goal of reducing GHG emissions. The World Bank and regional development banks, meanwhile, can play a key role in screening public and private flows going into countries that are already attractive investment locations, as well as provide inducements that reduce the risk of investors entering new markets in parts of Asia, Africa, and Latin America that have not received such flows to date.

What debates about governing climate finance often overlook is the need to de-finance the drivers of climate change. They often employ a "deficit model" focused on the staggering volumes of capital that need to be mobilized to decarbonize the global economy and effectively address the climate crisis. This focuses attention on the new infrastructures, financial products and services, technologies, and projects that will serve as key outlets for finance. What is often obscured, however, is the "equally important need to divest, redirect and withdrawal capital from climate change inducing fossil fuel infrastructures and patterns of growth."[36]

Who bears the costs?

In justice and equity terms, it is necessary to relate the question of who pays for action in financial terms to the question of who bears the costs of climate change, not just in terms of biophysical impacts, but also the human and social costs of how the issue is currently governed, or conversely, "ungoverned."

The findings contained in the last IPCC Assessment Report on the impacts of climate change demonstrate quite clearly that poorer and marginalized communities in drought-prone areas, those experiencing water scarcity, or those whose livelihoods depend on agriculture will be the worst affected and have the least capacity to adapt to climate change. As shown in Box 2.1, changes in the supply of water, natural resources, and food bring in their wake enormous social changes and disruptions. Across developing countries, it is areas of sub-Saharan Africa where poverty is most acute that will be worst affected. Indeed, even 20 years ago a multi-donor report on *Poverty and Climate Change* warned that "Climate change is a serious risk to poverty reduction and threatens to undo decades of development efforts."[37] As noted in the introduction, lack of progress in combating climate change will make it very difficult to realize most of the SDGs, just as the way in which SDGs concerning energy, food, sustainable cities, production, consumption, work, and growth are pursued will have a huge impact on the

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viability of climate goals. This requires us to think about the effectiveness and justice of climate governance in relation to these other domains of governance and politics.

BOX 2.1 CLIMATE IMPACTS ON FOOD AND WATER SECURITY

The most recent IPCC assessment report concluded:

- Climate change including increases in frequency and intensity of extremes have reduced food and water security, hindering efforts to meet Sustainable Development Goals
- Ocean warming and ocean acidification have adversely affected food production from shellfish aquaculture and fisheries in some oceanic regions
- Increasing weather and climate extreme events have exposed millions of people to acute food insecurity and reduced water security, with the largest impacts observed in many locations and/or communities in Africa, Asia, Central and South America, Small Islands and the Arctic
- Sudden losses of food production and access to food compounded by decreased diet diversity have increased malnutrition in many communities, especially for Indigenous Peoples, small-scale food producers and low-income households, with children, elderly people and pregnant women particularly impacted.
- Roughly half of the world's population currently experience severe water scarcity for at least some part of the year due to climatic and non-climatic drivers

Source: IPCC Sixth Assessment Report 2022, Summary for policymakers.

Given the slow pace of progress of international cooperation on climate change, noted in Chapter 1, and the difficulties of advancing some of the mitigation options discussed above, amid conflicts about how and by whom they should be governed, attention has increasingly turned to the issue of adaptation to climate change. Vulnerability to climate change cuts across geographical boundaries, and it is often the poor in all parts of the world that are affected most, raising issues of distributive justice.[38] The increasing intensity of climate-induced disruption around the world has

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revealed how marginalized communities bear some of the worst devastation inflicted by "natural" disasters magnifying as they do existing forms of social inequality on racial, class, and gender lines, for example.[39] Poorer people often have the least means of adapting their livelihoods and means of survival to sudden change of the sort associated with extreme weather events. Bradley Parks and Timmons Roberts argue, for example, that hurricane Mitch in Honduras "serves as a parable about uneven vulnerability to global climate change."[40] The Economic Commission for Latin America and the Caribbean (ECLAC) estimated that US$5 billion would be necessary to fund reconstruction efforts, to say nothing of damage to the banana plantations that were so central to generating export revenue that fell to one-fifth of their pre-hurricane levels, forcing Dole and Chiquita to lay off 25,000 workers and setting in train a series of other devastating social consequences. Likewise, the torrential rains and tropical cyclones which struck Mozambique and surrounding countries in 1999 left 700 people dead, one million people displaced, and extensive losses of land, animals, and other means of survival leaving a reconstruction bill of US$700 million.[41] Recent instances of heat waves in Europe and South Asia, wildfires in the United States, droughts in Australia, and deadly flooding in Thailand and Pakistan have further drawn attention to the devastating effects of climate change. Constructing effective mechanisms to support adaptation and pay for loss and damage raises issues not just of distributive, but also of procedural justice.[42]

The concern with issues of procedural justice reflects the fact that the impacts are felt and costs borne by poorer groups in particular, not just as a result of climate change and differential ability to adapt to it, but because of the ways in which mitigation and adaptation policies are designed and implemented.[43] Though the UN climate change negotiations may seem far removed from local land conflicts, the governance of climate change intersects with the governance of other key resources such as water and land.[44] For example, there have been cases of negative impacts on the poor when land is given up for carbon sink purposes, giving rise to accusations of carbon colonialism. In the case of the Clean Development Mechanism (CDM), REDD+ specifically and carbon forestry more broadly, this dynamic has played out in conflicts in places as diverse as Uganda, Mexico and Bolivia, and Southeast Asia.[45] Likewise, carbon finance can provide a financial lifeline to landfill sites which poorer communities had been campaigning to close down, but whose life is extended because of the profit to be made from capturing and burning methane.[46]

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People do not meekly accept their fate as victims of forms of climate governance they consider unjust, however. There is significant resistance to attempts at "accumulation by de-carbonization" by communities and movements resisting the commodification and inequities produced by "green grabs" of their land.[47] NGO watchdogs such as Carbon Trade Watch and Carbon Market Watch have also played an important role in monitoring public and private activities in the new carbon economy and drawing attention to the inequities it produces or fails to address, while REDD Monitor and the Indigenous Environment Network build networks of resistance to projects negatively impacting on the rights of indigenous peoples in particular.

In the search for means of protecting poorer groups from the effects of climate change or the effects of poorly conceived forms of climate governance, many affected communities and activists have increasingly turned to the language and tools of human rights. As Kevin Watkins of the United Nations Development Programme (UNDP) said of climate change: "it is about social justice and the human rights of the world's poor and marginalized. Failure to act on climate change would be tantamount to a systematic violation of the human rights of the poor."[48] For this reason, major human rights NGOs such as Amnesty International have come out in support of proposals for a fossil fuel non-proliferation treaty to regulate the supply of fossil fuels. As a mechanism for seeking redress for rights violations, access to the law is limited for many poorer groups, but there have been a series of legal cases which seek to explore the boundaries of culpability and establish new and challenging legal precedents.[49]

One interesting source of momentum for rights-based approaches to tackling climate injustices has come from groups adopting a range of legal-based strategies to hold governments and corporations to account for their obligations to act on the issue.[50] The landmark youth climate lawsuit against the US government (Juliana v. the United States) is a case in point; the Dutch court's order to the oil company Shell to achieve a specific emission reduction target along its entire supply chain is another,[51] while cases in France against the Macron government and in Norway and the Netherlands have sought to hold governments to account for their legal obligations to address climate change and prevent further climate injustice.

Civil society actors North and South also have a key role to play in debates about equity and justice in climate governance through enhancing the voice and reach of excluded groups. This can take the form of legal groups helping to redress some of the procedural inequities which mean levels of participation and effective engagement from smaller developing

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countries in the climate negotiations are low. Outside the formal negotiating arenas of climate governance, it includes youth movements such as Fridays for Future articulating intergenerational demands for climate justice and the climate justice movement which draws attention to disproportionate impact of climate change on the poor. This movement focuses on the way in which climate change has the potential to exacerbate existing social inequalities and draws much of its critique from broader contestations of neoliberal globalization. For the activists spearheading the movement, Climate Justice means

> holding fossil fuel corporations accountable for the central role they play in contributing to global warming … challenging these companies at every level – from the production and marketing of fossil fuels themselves, to their underhanded political influence, to their PR prowess, to the unjust "solutions" they propose, to the fossil fuel based globalization they are driving.[52]

The movement works through popular education and protest and seeks to provide a space for the articulation of claims by those most affected by climate change, but who contribute least to the problem. It has also spawned alliances such as the "Stop Climate Chaos" coalition involving environment and development NGOs working on climate justice issues and wanting to draw attention to impacts on the poor in particular,[53] as well as new waves of nonviolent direct action from movements such as Extinction Rebellion, 'Just stop oil' and forms of "blockadia" throughout the world seeking to call time on the fossil fuel economy and the governments that continue to support it.[54] We return to the issues raised by civic engagement in climate governance in Chapter 4.

Governance issues and challenges

From this discussion of the politics of responsibility, costs, and burdens of climate change, we find that there are a number of cross-cutting governance issues and challenges that arise. First, battles over the definition of the nature of the problem of climate change affect *how* and *where* it is addressed. In this chapter, we have seen how action on climate change has been presented as an investment opportunity by some (and as a loss by others), and how it has got caught up in, and at the same time redefines, older debates about aid and development. As connections to other issues and policy arenas are made, new actors are enrolled in climate governance

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from trade and financial institutions to bodies dealing with human rights and development. As bodies which ostensibly govern many of the key drivers of climate change in flows of finance and trade in carbon-intensive sectors, change in the mandates and direction of key global and regional economic governance bodies is in many ways a prerequisite to accelerated and more effective climate action. The new resources being made available to tackle climate change attract international organizations, civil society groups, and business alike. Carbon entrepreneurs seek a share of the market, while a growing body of NGOs define for themselves a role in addressing climate change and international agencies, from UNDP to the United Nations Industrial Development Organization (UNIDO), line up alongside the World Bank to make sure their presence is felt and their expertise brought to bear.

Just as each of these actors seeks to define the nature of the problem in a way which advances their interests, so too they compete to define appropriate solutions. In particular, the privileging of market-based solutions reflects the interests of particular organizations and nation-states with their neoliberal emphasis on using the market and allocating property rights to create incentives to reduce emissions wherever it is cheapest to do so. The proliferation of actors and sites of climate governance also raises significant challenges in terms of policy coordination and coherence, which in turn have implications for how climate action can be accelerated. Instead of coordinated strategies between global bodies working in relevant areas and across levels of governance from local government up to the global, we find high levels of incoherence. The activities of one body or set of organizations can systematically undermine those of others. New trade agreements increase the transport of goods over longer distances, adding to the emissions that climate negotiators are struggling to reduce,[55] investor dispute mechanisms are being invoked to contest governments' ambitions to phase out fossil fuels,[56] and multilateral development bank-lending continues to support projects that commit vast amounts of GHGs to the atmosphere.

We have seen in this chapter that whereas once climate change was an issue mainly addressed by the UN, its nature as a political issue at the intersection of the relationship between economy-energy and environment means that it connects to many other policy areas which in turn shape the effectiveness of action on climate change. This close interweaving of climate change with other key sectors of the economy, policy domains, and ecosystems raises another important issue—that of the "un-governance" of climate change: issues and areas of active and deliberate neglect. These are areas of politics and practice which are often not considered as climate

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policy, but which contribute significantly to the problem around energy, transport, agriculture, and trade, for example. This requires us to look not only at the activities of governments, but at the international institutions and market actors that shape so strongly the direction of national development strategies and, as a result, emissions trajectories.

Second, this chapter has shown how new sites for climate governance become venues for the articulation of demands that it has not been possible to achieve by other means or in other forums. The forging of links to aid, trade, and increasingly debt issues illustrates this clearly. For example, the average debt-to-GDP ratio in Africa is expected to increase to more than 70%, up from 60% in 2019, while most African countries could lose 2–4% of gross domestic product per year due to climate change by 2040. Recognizing this connection led the "Vulnerable Twenty" group to join others at COP26 in calling for an official mechanism to support an ambitious debt-for-climate-action swap whereby a creditor allows the debt to be reduced, with the money saved committed to poverty-reducing climate initiatives. For example, a swap in the Seychelles led to a US$27 million (approximately CAD 34 million) investment in fishery management, biodiversity conservation, and ecotourism.[57]

While some have sought to maintain clear boundaries around climate governance for fear of over-loading the agenda with the politics of these other issues and hence slowing progress, these new sites for climate governance also open up new potential avenues for advancing action. While the interlinkages between different policy issues create a challenge in terms of the governability of the "regime complex" around climate change, positive benefits might be achieved from acting on such synergies.[58] The emergence of new sites of climate politics is not, however, just about constructing new avenues of cooperation. Older development concerns re-emerge in the climate context. Concerns about subsidies to flailing technology sectors whose applications are inappropriate to contexts for which they were not designed and the use of non-additional tied aid which requires recipients to purchase the technology from the donor have all re-surfaced as concerns expressed by developing countries in the climate change arena.

Third, we find that issues of process have been decisive in shaping climate governance outcomes. Who gets to define issues of responsibility, equity, and justice is a function of who participates, who is represented, and who wields most power. Many smaller developing countries have limited representation in international negotiations, while many more powerful developing countries have moved to bilateral or pluri-lateral forums

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and "clubs" where they expect higher returns from their engagement and can avoid the complexities of multilateral forums. As climate governance enrolls more actors, questions of accountability come to the fore—whether it is the accountability of carbon market project developers to communities expected to host projects, the accountability of bodies such as the Green Climate Fund for the funds it disperses, or compensation for vulnerable areas of the world for climate impacts through loss and damage. This creates enormous challenges for the design of institutions that are transparent and open to participation, yet at the same time robust, effective, and proportionate to the scale of the climate crisis. Indeed, the need to enhance the effectiveness of action across scales and improve the accountability of key actors in climate governance has provided the point of departure for many of the transnational networks that are the subject of the next chapter.

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